CADETX VIRTUAL WORK EXPERIENCE

SPRINT 1 — CONSOLIDATED BUSINESS ANALYSIS QUESTIONS

Project: Heavy Supplier, Inventory & Warehouse Analytics
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1. Purpose

This document presents a consolidated set of 20 proposed business analysis questions for the Heavy Supplier, Inventory & Warehouse Analytics project.

The questions are designed to support business decision-making across sales performance, customer behaviour, product performance, inventory management, supplier performance, purchasing, financial management, and warehouse operations.

They were developed using the available warehouse table names and column definitions, together with the Sprint 1 KPI Proposal (V1).

The questions are intended for team review and prioritisation. Their inclusion does not imply that the analysis has already been performed or that the results have been validated.

2. Business Analysis Questions

ID	Focus Area	Business Analysis Question	Relevant Tables
BA-01	Sales Performance	Which branches generate the highest delivered sales value?	`branches`, `sales_orders_header`
BA-02	Sales Performance	What are the monthly sales trends for delivered orders?	`sales_orders_header`
BA-03	Sales Performance	Which products generate the highest sales revenue?	`products`, `sales_orders_header`, `sales_orders_lines`
BA-04	Sales Performance	Which branches have the highest order cancellation rates?	`branches`, `sales_orders_header`

BA-05	Customer Analysis	Which customers have the highest total purchase value?	`customers`, `sales_orders_header`
BA-06	Customer Analysis	Which customer segments contribute the most to sales revenue?	`customers`, `sales_orders_header`
BA-07	Customer Analysis	Which customers have outstanding balances requiring follow-up?	`customers`, `invoices`, `payments`
BA-08	Customer Analysis	Which regions contribute the most to customer purchases?	`customers`, `sales_orders_header`
BA-09	Product Analysis	Which product categories generate the highest sales revenue?	`products`, `sales_orders_lines`, `sales_orders_header`
BA-10	Product Analysis	Which products have the highest profit margins based on their recorded costs and selling prices?	`products`
BA-11	Inventory Management	Which products are below their reorder levels across branches?	`inventory_master`, `products`, `branches`
BA-12	Inventory Management	Which products have stock levels below their defined safety-stock levels?	`inventory_master`, `products`, `branches`
BA-13	Inventory Management	Which products and branches experience the most frequent stock movements?	`stock_ledger`, `products`, `branches`
BA-14	Supplier Performance	Which suppliers have the longest recorded delivery lead times?	`suppliers`, `purchase_orders_header`
BA-15	Supplier Performance	Which suppliers have the lowest reliability scores?	`suppliers`
BA-16	Purchasing Analysis	Which suppliers account for the highest total purchasing costs?	`suppliers`, `purchase_orders_header`, `purchase_orders_lines`

BA-17	Purchasing Analysis	Which purchase orders were received later than their expected delivery dates?	`purchase_orders_header`, `suppliers`
BA-18	Warehouse Operations	Which branches have the highest average monthly revenue relative to their monthly operational costs?	`branches`
BA-19	Financial Analysis	Which invoices remain unpaid or have outstanding balances?	`invoices`, `payments`, `customers`
BA-20	Financial Analysis	How does recorded payment activity compare with the value of invoiced sales?	`invoices`, `payments`

3. Key Business Objectives

The proposed questions aim to help the business:

1. **Improve sales performance:** Identify high-performing branches, products, and customer segments.
2. **Strengthen customer management:** Understand customer purchasing patterns and identify balances requiring follow-up.
3. **Optimise inventory:** Identify products that may require replenishment and monitor stock movements.
4. **Evaluate supplier performance:** Assess supplier reliability and delivery timeliness.
5. **Improve purchasing decisions:** Understand purchasing costs and identify delayed purchase orders.
6. **Monitor financial performance:** Examine invoiced sales, recorded payments, and outstanding balances.
7. **Evaluate warehouse operations:** Compare branch revenue and operational costs to identify areas for further investigation.

4. Data Requirements and Validation Considerations

Before conducting the analysis, the team should confirm that the relevant fields contain valid, consistent, and sufficiently complete data.

The following considerations are important:

- **Sales analysis:** Use the agreed sales-value field and confirm whether only delivered orders should be included in sales performance metrics.
- **Customer analysis:** Confirm customer identifiers and relationships between customer records and sales orders.

- **Inventory analysis:** Validate stock quantities and confirm the relationship between product and branch identifiers.
- **Supplier analysis:** Distinguish recorded supplier lead times from actual delivery delays calculated using expected and received delivery dates.
- **Purchasing analysis:** Confirm purchase-order statuses and validate the calculation of purchase costs.
- **Financial analysis:** Investigate repeated invoice and payment identifiers before relying on invoice-payment reconciliation results.
- **Warehouse operations:** Treat revenue-to-cost comparisons as preliminary indicators and account for the reporting period and any relevant operational differences between branches.

The original source data should be preserved. Any cleaning, deduplication, or approved business-rule changes should be documented according to the project's data-quality procedures.

5. Expected Outcome

The consolidated questions provide a starting point for identifying the business insights required from the warehouse.

Following team review, the questions can be prioritised according to business relevance, data availability, analytical feasibility, and stakeholder requirements. Approved questions can then guide KPI selection, SQL analysis, and dashboard development.

Document Status: Submitted for team review and refinement.